

A Note on Equation (364) in Dimock's *The Renormalization Group According to Balaban, I. Small Fields*

July 6, 2026

Conclusion

The estimate printed as Eq. (364),

$$|C_k(\Omega; y, y')| \leq O(1)L^2 \exp\{-\delta_0 L^{-2} d(y, y')\}, \quad (364)$$

is a valid weaker bound, but the factor L^{-2} is not the factor produced by the scaling argument given immediately before it. Scaling Eq. (348) from $G_{k+1}(L^{-1}\Omega)$ to $G_{k+1}^0(\Omega)$ gives instead

$$|\langle f, G_{k+1}^0(\Omega) f' \rangle| \leq CL^2 \exp\{-cL^{-1} d(y, y')\} \|f\|_2 \|f'\|_2, \quad (1)$$

and hence

$$|C_k(\Omega; y, y')| \leq CL^2 \exp\{-cL^{-1} d(y, y')\}. \quad (2)$$

Here $c > 0$ is independent of k, N, Ω, y, y' , and may be taken as a fixed constant multiple of the δ_0 in Eq. (348).

Since $L^{-1} \geq L^{-2}$ for $L > 1$, the sharp estimate (2) immediately implies the printed estimate (364), after possibly changing constants. Thus Eq. (364) is harmless for the later argument. However, if Eq. (364) is read as the estimate that directly follows from scaling, the exponent should contain L^{-1} , not L^{-2} . This is also confirmed internally by Eq. (370), whose first displayed line uses $\exp\{-\delta_0 L^{-1} d(y, y')\}$ while citing Eq. (364).

A further consequence is that the decay rate in Eq. (374) need not be of order L^{-2} . If one uses (2) sharply, then the final estimate for $G_k(\Omega)$ may be written with an L -independent exponential rate,

$$|(G_k(\Omega)f)(x)| \leq C_L \exp\{-\Gamma d(x, \text{supp } f)\} \|f\|_\infty, \quad \Gamma = O(1).$$

Here C_L may depend on L , as is allowed by the convention in the paper. If, however, one keeps the extra weakening by L^{-1} that appears in the printed Eq. (373), the resulting final rate is only $O(L^{-1})$. Thus the printed $O(L^{-2})$ in Lemma 33 is not forced by the improved Eq. (364).

The two substantive losses of sharpness are:

$$\text{Eq. (364): } L^{-1} \longrightarrow L^{-2}, \quad \text{Eq. (373): } \Gamma = L\alpha_L \longrightarrow \Gamma = \alpha_L.$$

The first loss is the replacement of the scaling-correct C_k decay rate by a weaker one. The second loss is the extra L^{-1} inserted when Lemma 32 is used inside the multiscale identity (372).

Notation

In Appendix D, Dimock considers

$$G_k(\Omega) = [-\Delta + \bar{\mu}_k + a_k Q_k^T Q_k]_{\Omega}^{-1}, \quad (340)$$

on $\Omega \subset (L^{-k}\mathbb{Z})^3$. Equation (348) states that if $\text{supp } f \subset \Delta_y$, $\text{supp } f' \subset \Delta_{y'}$, then

$$|\langle f, G_k(\Omega) f' \rangle| \leq C \exp\{-\delta_0 d(y, y')\} \|f\|_2 \|f'\|_2. \quad (348)$$

The operator relevant for Eq. (363) is

$$G_{k+1}^0(\Omega) = [-\Delta + \bar{\mu}_k + L^{-2} a_{k+1} Q_{k+1}^T Q_{k+1}]_{\Omega}^{-1}. \quad (362)$$

This is the unscaled $k+1$ Green function acting on the old lattice $(L^{-k}\mathbb{Z})^3$. It becomes the ordinary G_{k+1} after the spatial change of variables $x = Lz$.

Scaling of G_{k+1}^0

Let

$$\Omega' = L^{-1}\Omega \subset (L^{-(k+1)}\mathbb{Z})^3$$

and define the pullback

$$(Rf)(z) = f(Lz), \quad z \in \Omega'.$$

With the usual weighted lattice sums, the measure scales by $dx = L^3 dz$, so

$$\|Rf\|_{L^2(\Omega')} = L^{-3/2} \|f\|_{L^2(\Omega)}. \quad (3)$$

The lattice Laplacian scales as

$$R(-\Delta_k)R^{-1} = L^{-2}(-\Delta_{k+1}), \quad (4)$$

and the mass satisfies

$$\bar{\mu}_k = L^{-2} \bar{\mu}_{k+1}. \quad (5)$$

Moreover the averaging term in G_{k+1}^0 is precisely the old-lattice representation of the $k+1$ averaging term after scaling:

$$R(L^{-2} Q_{k+1}^T Q_{k+1}) R^{-1} = L^{-2} Q_{k+1}^T Q_{k+1}. \quad (6)$$

The same symbol Q_{k+1} appears on both sides, but the left side acts on $\Omega \subset (L^{-k}\mathbb{Z})^3$, whereas the right side acts on $\Omega' \subset (L^{-(k+1)}\mathbb{Z})^3$.

Combining (4), (5), and (6),

$$R(G_{k+1}^0(\Omega))^{-1} R^{-1} = L^{-2} (G_{k+1}(\Omega'))^{-1}.$$

Therefore

$$RG_{k+1}^0(\Omega)R^{-1} = L^2 G_{k+1}(\Omega'). \quad (7)$$

Equivalently,

$$G_{k+1}^0(\Omega)f = L^2 R^{-1} G_{k+1}(\Omega') Rf.$$

Now compute the L^2 pairing. For any u on Ω and v on Ω' ,

$$\langle u, R^{-1}v \rangle_{\Omega} = L^3 \langle Ru, v \rangle_{\Omega'}.$$

Using (7),

$$\begin{aligned} \langle f, G_{k+1}^0(\Omega)f' \rangle_{\Omega} &= L^2 \langle f, R^{-1}G_{k+1}(\Omega')Rf' \rangle_{\Omega} \\ &= L^5 \langle Rf, G_{k+1}(\Omega')Rf' \rangle_{\Omega'}. \end{aligned} \quad (8)$$

Apply Eq. (348) to $G_{k+1}(\Omega')$. The supports of Rf and Rf' are the scaled sets $L^{-1}\Delta_y$ and $L^{-1}\Delta_{y'}$. Their separation is

$$\text{dist}(L^{-1}\Delta_y, L^{-1}\Delta_{y'}) = L^{-1}d(y, y') + O(1),$$

where the $O(1)$ ambiguity only changes the prefactor. Hence

$$|\langle Rf, G_{k+1}(\Omega')Rf' \rangle_{\Omega'}| \leq C \exp\{-cL^{-1}d(y, y')\} \|Rf\|_2 \|Rf'\|_2.$$

Using (3) in (8) gives

$$\begin{aligned} |\langle f, G_{k+1}^0(\Omega)f' \rangle_{\Omega}| &\leq L^5 C e^{-cL^{-1}d(y, y')} (L^{-3/2} \|f\|_2) (L^{-3/2} \|f'\|_2) \\ &= CL^2 e^{-cL^{-1}d(y, y')} \|f\|_2 \|f'\|_2. \end{aligned}$$

This proves (1). Thus the exponent obtained from scaling is $L^{-1}d(y, y')$, not $L^{-2}d(y, y')$.

From G_{k+1}^0 to C_k

Equation (361) is

$$C_k(\Omega) = A_k + a_k^2 A_k Q_k G_{k+1}^0(\Omega) Q_k^T A_k, \quad (361)$$

where

$$A_k = [a_k + aL^{-2}Q^T Q]_{\Omega}^{-1}.$$

Since $Q^T Q$ is a projection,

$$A_k = a_k^{-1}(I - Q^T Q) + (a_k + aL^{-2})^{-1}Q^T Q.$$

Thus A_k is local, with uniformly bounded operator norms and bounded local range. The operators Q_k and Q_k^T are local averaging and extension operators. Consequently, $A_k Q_k$ and $Q_k^T A_k$ only smear a delta function over a bounded neighborhood of the corresponding unit block, and their L^2 norms are uniformly bounded.

For the nonlocal term in (361), apply (1) to

$$f = Q_k^T A_k \delta_y, \quad f' = Q_k^T A_k \delta_{y'}.$$

The supports of these functions are within bounded distance of Δ_y and $\Delta_{y'}$, respectively, and their L^2 norms are $O(1)$. Therefore

$$|(A_k Q_k G_{k+1}^0(\Omega) Q_k^T A_k)(y, y')| \leq CL^2 e^{-cL^{-1}d(y, y')}.$$

The first term $A_k(y, y')$ is local. If it is nonzero, then $d(y, y') = O(1)$, and it is also bounded by the same right side after increasing C . Hence

$$|C_k(\Omega; y, y')| \leq CL^2 e^{-cL^{-1}d(y, y')},$$

which is (2).

Consequence for Lemma 32

Now insert the improved estimate (2) into the proof of Lemma 32. Recall

$$C'_k(\Omega) = H_k(\Omega)C_k(\Omega)H_k(\Omega)^T.$$

Lemma 31 gives, with $\delta_1 = O(1)$,

$$|H_k(\Omega; x, y)| \leq Ce^{-\delta_1 d(x, y)}, \quad |H_k(\Omega; y', z)| \leq Ce^{-\delta_1 d(y', z)}.$$

Thus, for $A = \text{supp } f$,

$$|(C'_k(\Omega)f)(x)| \leq C_L \sum_{y, y'} e^{-\delta_1 d(x, y)} e^{-cL^{-1}d(y, y')} e^{-\delta_1 d(y', A)} \|f\|_\infty. \quad (9)$$

The prefactor C_L absorbs the L^2 in (2) and the summability constants. Since

$$d(x, A) \leq d(x, y) + d(y, y') + d(y', A),$$

the convolution in (9) decays at the slowest of the three displayed rates. Therefore

$$|(C'_k(\Omega)f)(x)| \leq C_L e^{-\alpha_L d(x, \text{supp } f)} \|f\|_\infty, \quad \alpha_L = c_0 L^{-1}, \quad (10)$$

with $c_0 = O(1)$ independent of L . The same reasoning gives the derivative and Holder-derivative estimates, since the corresponding bounds on H_k have the same $O(1)$ decay rate δ_1 .

Thus the improved Eq. (364) improves Lemma 32 from the paper's $\gamma_0 = O(L^{-2})$ to an intermediate one-step rate $O(L^{-1})$. It does not make Lemma 32 itself $O(1)$, because C_k still has only $e^{-cL^{-1}d}$ decay.

Consequence for Eq. (374)

The key point is that Eq. (372) contains an additional scaling by L^{k-j} . Write $n = k - j$, so $1 \leq n \leq k$, and put

$$D = d(x, \text{supp } f).$$

The identity (372) is

$$(G_k(\Omega)f)(x) = \sum_{j=0}^{k-1} L^{-2(k-j)} (C'_j(L^{k-j}\Omega)f_{L^{k-j}})(L^{k-j}x). \quad (372)$$

Equivalently,

$$(G_k(\Omega)f)(x) = \sum_{n=1}^k L^{-2n} (C'_{k-n}(L^n\Omega)f_{L^n})(L^n x).$$

The support rescales as

$$d(L^n x, \text{supp } f_{L^n}) = L^n d(x, \text{supp } f) = L^n D.$$

Applying (10) directly gives

$$|(G_k(\Omega)f)(x)| \leq C_L \sum_{n=1}^k L^{-2n} e^{-\alpha_L L^n D} \|f\|_\infty. \quad (11)$$

Since $\alpha_L = c_0 L^{-1}$,

$$\alpha_L L^n = c_0 L^{n-1} \geq c_0 \quad (n \geq 1).$$

Hence each term in (11) satisfies

$$e^{-\alpha_L L^n D} \leq e^{-c_0 D},$$

and

$$\sum_{n=1}^k L^{-2n} \leq \sum_{n=1}^{\infty} L^{-2n} \leq C.$$

Therefore the final Green-function estimate can be written as

$$|(G_k(\Omega)f)(x)| \leq C_L e^{-c_0 d(x, \text{supp } f)} \|f\|_{\infty}. \quad (12)$$

The decay rate c_0 is $O(1)$, independent of L . The prefactor may depend on L , in agreement with the paper's convention that constants denoted C may depend on L .

This conclusion uses (10) sharply. The printed Eq. (373) contains an additional weakening:

$$e^{-\gamma_0 d(L^n x, \text{supp } f_{L^n})} \quad \text{is replaced by} \quad e^{-\gamma_0 L^{-1} d(L^n x, \text{supp } f_{L^n})}.$$

If one makes the analogous weakening after improving Lemma 32, then

$$e^{-\alpha_L L^{-1} d(L^n x, \text{supp } f_{L^n})} = e^{-c_0 L^{n-2} D},$$

and the first term $n = 1$ gives only $e^{-c_0 L^{-1} D}$. In that weakened reading the final rate is $O(L^{-1})$, not $O(1)$.

Where Sharpness Is Lost

It is useful to separate the rates in the three objects

$$C_k, \quad C'_k = H_k C_k H_k^T, \quad G_k.$$

Let

$$\beta_L = \text{decay rate in } C_k, \quad \alpha_L = \text{decay rate in } C'_k, \quad \Gamma_L = \text{final decay rate in } G_k.$$

The sharp propagation of rates is

$$\beta_L \implies \alpha_L \simeq \min\{\delta_1, \beta_L\} \implies \Gamma_L \simeq L \alpha_L.$$

The first implication is just the convolution with the two H_k kernels. Since the H_k kernels decay at the $O(1)$ rate δ_1 , they cannot improve a slower C_k rate; they preserve it up to harmless constants. The second implication comes from the factor

$$d(L^n x, \text{supp } f_{L^n}) = L^n d(x, \text{supp } f)$$

in Eq. (372), and the worst case $n = 1$ gives the gain L .

Thus, if the improved Eq. (364) is used, then

$$\beta_L \simeq L^{-1}, \quad \alpha_L \simeq L^{-1}, \quad \Gamma_L \simeq L \alpha_L \simeq O(1).$$

The first non-sharp step is the printed scaling estimate (363), and hence the printed Eq. (364). Scaling Eq. (348) from $G_{k+1}(L^{-1}\Omega)$ gives

$$|\langle f, G_{k+1}^0(\Omega)f' \rangle| \leq CL^2 e^{-cL^{-1}d(y,y')} \|f\|_2 \|f'\|_2,$$

because distances are divided by L under $\Omega \mapsto L^{-1}\Omega$. The printed $e^{-\delta_0 L^{-2}d(y,y')}$ is therefore a weakening by one power of L . Passing from G_{k+1}^0 to C_k through

$$C_k = A_k + a_k^2 A_k Q_k G_{k+1}^0 Q_k^T A_k$$

does not force any further loss, because A_k, Q_k, Q_k^T are local. Their support enlargement is $O(L)$, and

$$e^{-cL^{-1}(d-O(L))} \leq C e^{-cL^{-1}d}.$$

The second non-sharp step is Eq. (373). If Lemma 32 gives

$$|(C'_j F)(X)| \leq C_L e^{-\alpha_L d(X, \text{supp } F)} \|F\|_\infty,$$

then a sharp substitution into Eq. (372) gives

$$|(C'_{k-n}(L^n \Omega) f_{L^n})(L^n x)| \leq C_L e^{-\alpha_L L^n d(x, \text{supp } f)} \|f\|_\infty.$$

The printed Eq. (373) instead replaces this by

$$C_L e^{-\alpha_L L^{-1}d(L^n x, \text{supp } f_{L^n})} \|f\|_\infty = C_L e^{-\alpha_L L^{n-1}d(x, \text{supp } f)} \|f\|_\infty.$$

This loses exactly one power of L in the final rate. It changes

$$\Gamma_L = L\alpha_L \quad \text{into} \quad \Gamma_L = \alpha_L.$$

Everything after this point is sharp relative to the weakened input; the geometric sum in Eq. (374) does not introduce a new decay loss.

The complete rate ledger is therefore:

input/procedure	final rate in Eq. (374)
printed Eq. (364), used sharply in Eq. (372)	$O(L^{-1})$
printed Eq. (364), plus printed Eq. (373) weakening	$O(L^{-2})$
improved Eq. (364), but printed Eq. (373) weakening	$O(L^{-1})$
improved Eq. (364), used sharply in Eq. (372)	$O(1)$

Why the Printed L^{-2} Is Harmless

The sharper estimate implies the printed estimate because, for $L > 1$ and $d \geq 0$,

$$e^{-cL^{-1}d} \leq e^{-cL^{-2}d}.$$

Thus

$$CL^2 e^{-cL^{-1}d(y,y')} \leq CL^2 e^{-cL^{-2}d(y,y')}.$$

This explains why the later proof can safely use a decay rate of order L^{-2} . Indeed Lemma 32 explicitly sets

$$\gamma_0 = \frac{1}{2} \delta_0 L^{-2}.$$

However, Eq. (370) begins with

$$e^{-\delta_1 d(x,y)} e^{-\delta_0 L^{-1} d(y,y')} e^{-\delta_1 d(y', \text{supp } f)},$$

while saying “By (364) and (366).” This line is incompatible with the printed form of Eq. (364), but it is exactly compatible with the scaled estimate (2). The most coherent reading is therefore:

- the estimate actually obtained by scaling, and apparently used in the first line of Eq. (370), has L^{-1} in the exponent;
- the weaker L^{-2} bound is sufficient for the subsequent argument and is the rate finally retained in Lemma 32 and Lemma 33;
- if the improved estimate is propagated sharply through Eq. (372), the final decay rate in Eq. (374) can be taken independent of L .